

James G. E. Ogilvie

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Profile

Evolutionary geneticist studying how new traits arise, using the diverse warning colour patterns of *Heliconius* butterflies as a model. My research combines regulatory and 3D genomics (Hi-C, ChIP-seq, ATAC-seq, RNA-seq) with population genomics to understand how gene regulation evolves, and how hybridization between colour pattern populations can bring regulatory switches together to create new phenotypes. This work is complemented by field experiments across Central and South America on the selective pressures that maintain warning colour diversity. Experienced in Python, R and HPC-based analysis, and in collaborating across computational and experimental teams.

Technical skills

Programming & data: Python, R, bash, Linux/HPC, reproducible analysis workflows.

NGS / multi-omics: RNA-seq, ATAC-seq, ChIP-seq, Hi-C; QC, alignment, quantification, differential analysis, peak-based analyses.

Genomics: Regulatory genomics, QC, analysis, interpretation, variant calling and genotyping (GATK); annotation-aware genomic analyses. Integration of multi-omic datasets to generate biologically coherent conclusions.

Scientific reasoning & evaluation: Construction and review of complex, real-world biology problem statements. Statistical reasoning and validation of biological claims. Experience collaborating asynchronously across computational and experimental teams.

Appointments & awards

Graduate Research Scholars Program Grantee

Alabama EPSCoR · 2022–2023

Pre-Doctoral Fellow

Smithsonian Tropical Research Institute, Panama · 2018–2022

Education

PhD, Evolutionary Genomics

Auburn University, USA · 2024

MSc, Zoology

Universidad Autónoma de Madrid, Spain · 2015

BSc, Ocean Science

Plymouth University, UK · 2013

Selected publications

Ogilvie, J.G., Lewis, J., ... Counterman, B.A. (In revision). Multi-locus diversification underlies the origin of a hybrid wing colour pattern.

Ogilvie, J.G. & Counterman, B.A. (2025). Hybrid speciation. In Kliman, R.M. (Ed.), *Encyclopedia of Evolutionary Biology* (2nd ed.). Elsevier.

Van Belleghem, S.M., ... **Ogilvie, J.G.**, ... (2023). High level of novelty under the hood of convergent evolution. *Science*, 379(6636), 1043–1049. doi:10.1126/science.ade0004

Ogilvie, J.G., Van Belleghem, S.M., Range, R., Papa, R., McMillan, W.O., Chouteau, M. & Counterman, B.A. (2021). Balanced polymorphisms and their divergence in a *Heliconius* butterfly. *Ecology and Evolution*, 11(24), 18319–18330.
doi:10.1002/ece3.8423

Selected talks

The selective pressures maintaining polymorphic mimicry in a Müllerian mimic

Southeastern Population Ecology and Evolutionary Genetics Conference (SEPEEG) · Awarded top graduate speaker

The evolution of polymorphic mimicry in *Heliconius* butterflies

Society for Integrative and Comparative Biology (SICB)

The evolutionary pressures and genetics of wing pattern variation in *Heliconius* butterflies

11th International Workshop on Molecular Biology and Genetics of the Lepidoptera (INRAE), Kolymari, Crete

Professional experience

Fundraising & Events Coordinator

PCFML, NSW, Australia · 2016–2018

Intern Ecologist

Centre for the Mexican Turtle, Oaxaca, Mexico · 2015

Surfing Coach

Star Surf Camps, France / Spain / Australia · 2015–2017

Translator (Spanish/English)

Sotogrande S.A., Cádiz, Spain · 2010–2013

Languages

English (native), Spanish (native), Portuguese (basic)

Interests

Artificial intelligence, nature and biodiversity, surfing

Referee

Dr Brian Counterman, PhD Advisor, Auburn University · bac0071@auburn.edu
Further references available on request.